



Project 5

Performance Tuning



Source Dataset 1



- Orders Management
- Format – CSV
- Total Data Size – 105 GB
- Total no. of files – 39
- Total queries – 18

```
order_id  
shipment_date  
invoice_number  
shipment_from_city  
shipment_to_city  
invoice_amount
```

Order Shipment

```
order_id  
order_date  
customer_id  
total_order_amount  
order_discount_pct  
final_order_amount  
customer_name  
customer_country  
customer_city  
customer_state  
customer_credit_rating  
customer_premium_flag
```

Order Transaction

```
fc_id  
fc_release_to_mfg_date  
fc_release_to_mfg_count  
fc_arrival_from_mfg_date  
fc_arrival_from_mfg_count  
fc_inspection_date  
fc_mfg_facility_id  
fc_release_to_shipment_date  
mfg_facility_id  
to_mfg_facility_arrival_date  
from_mfg_facility_departure_date  
mfg_facility_input_count  
mfg_facility_output_count  
mfg_facility_backlog_count  
mfg_facility_product_capability_count
```

Order Fulfillment

Set up



- Create 3 tables as follows:
 - Order_shipment – Dist Key: order_id, Primary Key: order_id, Dist Style: EVEN
 - Order_transaction - Dist Key: order_id, Primary Key: order_id, Dist Style: KEY
 - Order_fulfillment - Dist Key: fc_id, Primary Key: fc_id, Dist Style: KEY
- Load the tables using the files (COPY command).
- Check load status every 2 mins.
- Check the size of each table.
- Check distribution of slice.
- Confirm DISTKEY, SORTKEY and DISTSTYLE using metadata table.

Pre-tuning Step 1



- Perform dimensional modelling, if necessary.
- Load newly created tables from dimensional modelling, if it was done in step 1 above. Has to be done from table to table using SQL statements.
- Following are 1st set of queries:
 - Get all orders of a customer that are more than 2 yrs old and less than 8 yrs.
 - What is the range of discount that customers are getting. How are they related to credit rating.
 - Which country places the most orders. This must be retrieved from last years data.
 - What is the avg credit rating of customers country-wise.
 - What is the avg. shipment time in days for premium customers.
 - What is the avg. shipment time between all source and destination cities.

Pre-tuning Step 1



- Continuing 1st set of queries:
 - What is the avg. lead time between order and shipment. Lead time = order date - shipment date
 - Is there mismatch between customer city and source shipment city. If yes, fix it. Order Transaction is the source of truth.
 - Is the lead time related to order amount.
 - Find top 20 states that have the highest premium customers.
- Following are 2nd set of queries:
 - Sort manufacturing centers in order of best to least delivery time (depart dt – arrival dt).
 - Sort fulfillment centers in descending order of backlog.
 - FC backlog = for the same date (release to mfg. count – arrival from mfg. count)
 - Which FC has the highest inspection.

Pre-tuning Step 1



- Continuing 2nd set of queries:
 - What is the avg. delay between FC release date and mfg. facility arrival days.
 - Which FC have released the most orders last year and to which mfg. facility.
 - Match the reported backlog count (mfg_facility_backlog_count) as opposed to actual (mfg_facility_input_count - mfg_facility_output_count) backlog count for each mfg. facility.
 - List top 25 mfg. facility for their capability.
 - Is the backlog count related to diverse capabilities.

Pre-tuning Step 2



- Benchmark the queries.
 - Complete dimensional modelling.
 - Load the new tables.
 - Execute each query and note the timing.
 - Set SLA and priority.

Tuning



- Tune the queries:
 - Check EXPLAIN output. Check all operators, system joins etc.
 - Try to map EXPLAIN output with SVL_QUERY_SUMMARY.
 - Check Stream, Segment and Step for the queries to find bottleneck.
 - Run VACUUM, ANALYZE.
 - Use different system tables.
 - Change SORT KEY, DIST KEY, DIST STYLE etc.
 - Create different tables with different structure (distkey, sortkey etc.) to check query execution time.
- Once you decide with new structure perform the above exercise again.

Source Dataset 2



- Financial Data
- Format – CSV
- Total Data Size – 25 GB
- Total no. of files – 16
- Total queries – 10

```
ledger_id  
transaction_id  
account_id  
ledger_type  
amount  
entry_date
```

Ledger

```
loan_id  
customer_id  
loan_type  
principal_amount  
interest_rate  
term_month  
start_date  
end_date
```

Loan Account

```
account_id  
customer_id  
balance  
currency_code  
acc_opening_date
```

SB Account

```
transaction_id  
src_account_id  
dest_account_id  
amount  
currency_code  
transaction_date  
tran_status
```

Transaction

```
customer_id  
first_name  
last_name  
email  
phone_number  
date_of_birth  
gender  
national_id  
address_line_1  
address_line_2  
city  
country  
postal_code  
created_at  
updated_at
```

Customer

Business Statements



- Find the top 5 customers with most transactions in the last 5 years.
- How many customers that have taken loan already holds account.
- Find customers with account more than 5 years.
- Which currency has the highest failed and completed transactions.
- Find the top 10 transactions (in amount). Which cities those belong to.
- Which country spends the most on expenses. List top 25 highest income countries.
- Find country-wise loan distribution.
- Find the money in equity for individual countries.
- Which age group takes the highest loan and which type of loan.
- What is the most used currency internationally (based on ledger). Currency should match customer's country.



Thank you

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